Fiscal Year:	FY 2004	**Task Last Updated:**	FY 01/13/2005
PI Name:	Ciszak, Ewa M. Ph.D.		
Project Title:	Crystal Structure of Human Pyruvate Dehydrogenase complex Facilitated by Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Structural biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Huntsville	**State:**	AL
Zip Code:	5812	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-03
Start Date:	09/01/2001	**End Date:**	03/30/2005
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Patel, Mulchand (SUNY at Buffalo) Hong, Young (BAE Analytical)		
Grant/Contract No.:	NAG8-1924		
Performance Goal No.:			
Performance Goal Text:			

The proposed research falls in the category of the test of microgravity and the use of the International Space Station as a platform for research in the area of biotechnology and design of macromolecular materials. The research involves crystallization of component proteins of human pyruvate dehydrogenase complex using the terrestrial and low gravity environments followed by the analyses of X-ray diffraction data and structural determinations. These efforts should allow determination of benefit of microgravity for improving or not crystal quality for hPDC. Human pyruvate dehydrogenase complex (hPDC) is an intramitochondrial multienzyme complex consisting of multiple copies of several subunits including three enzyme activities E1 (pyruvate dehydrogenase), E2 (dihydrolipoamide acetyltransferase) and E3 (dihydrolipoamide dehydrogenase), required for the completion of the conversion of pyruvate to acetyl CoA, and E3BP (E3-binding protein) required for binding E3 component enzyme in this protein assembly. E1 catalyses the non-reversible removal of carbon dioxide from pyruvate; E2 forms acetyl CoA and E3 reduces NAD+ to NADH. Two

Task Description:	additional enzyme activities are associated with the complex: a specific kinase which is capable of phosphorylating E1 at three serine residues and a loosely-associated specific phosphatase which reverses the phosphorylation The proportion of the PDC in its active dephosphorylated state is determined by a balance between the activity of the kinase and phosphatase. The activity of the kinase may be regulated in vivo by the relative concentrations of metabolic substrates such as pyruvate, lipoic acid covalently bound to E2, NAD+/NADH, CoA/acetylCoA as well as adenine diphosphate (ADP)/ATP. We propose to carry out X-ray crystallographic studies of component proteins of arguably the most challenging multienzyme assembly of the human body. Our research will focus on obtaining crystals of component proteins of the core of hPDC including E1, E2-binding fragment and E3, and determination of their structures. Crystallization conditions will also be tested in low gravity environment to demonstrate its value in improving crystal quality. We will also take care to construct validation protocols that should allow us to determine the impact of microgravity on the progress of our structural studies. Our research is designed to (1) validate the advantage of low-gravity as a tool for improving crystal quality and (2) achieve the structure of proteins of arguably the most challenging multienzyme assembly, human pyruvate dehydrogenase complex.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	N/A
Task Progress:	Achievements (09/2003-09/2004). 1. Investigators prepared publications and presentations related to determination of structures of pyruvate dehydrogenase and novel procedures for use in delayed crystallization (see Bibs, Patents, Awards). 2. Several X-ray diffraction data sets were analyzed for the structure determination of pyruvate dehydrogenase component of hPDC. The structure of pyruvate dehydrogenase and its mechanism of catalytic action have been extended to computational analyses of variants associated with genetic diseases. 3. The PI provided training and advice to summer interns supporting this project in some extend.
Bibliography Type:	Description: (Last Updated: 01/13/2005)
Articles in Peer-reviewed Journals	Korotchkina, L.G.; Ciszak, E.M.; Patel, M.S. "Function of several critical amino acids in human pyruvate dehydrogenase revealed by its structure" Archives of Biochemistry and Biophysics, 2004, 429, 171-179 , Sep-2004
Articles in Peer-reviewed Journals	Potter, R.R.; Hong, Y.S.; Ciszak, E.M. "Use of plastic capillaries for macromolecular crystallization" Journal of Applied Crystallography, 2004, 37, 500-501. , May-2004
Articles in Peer-reviewed Journals	Ciszak, E.M.; Korotchkina, L.G.; Dominiak, P.M.; Sidhu, S.; Patel, M.S. "Structural basis for flip-flop action of thiamin pyrophosphate-dependent enzymes revealed by human pyruvate dehydrogenase" Journal of Biological Chemistry, 2003, 278, 21240-21246. , Jun-2003
Presentation	Ciszak, E.M.; Dominiak, P.M. "The Thiamin Pyrophosphate Action -motif" Frontiers in Structural Biology, Keystone Symposia, Abstract Book, p.122, Snowbird UT. Apr-2004
Presentation	Makal, A.; Hong, Y.S.; Potter, R.R.; Vettaikkorumakankauv, A.K.; Korotchkina, L.G.; Patel, M.S.; Ciszak E.M. "Intricate crystal structure of dihydrolipoamide dehydrogenase with its binding protein: multiple copies, dynamic and static disorders " Annual American Crystallographic Meeting, Program and Abstract Book, p.92, Chicago, IL Jul-2004